

Major Project Documentation

Week 2 – Field Visit Discussion Report

Location: AIISH (All India Institute of Speech and Hearing), Mysore

Discussion with: Dr. Yashaswini

Date: 11 March 2026

Discussion from Second Visit

Subject: Advanced Therapeutic Modalities for Post-Stroke Recovery

- **The Expressive Paradox:** Dr. Ashwini highlighted a common clinical phenomenon where patients maintain high levels of auditory comprehension but suffer from a total or partial breakdown in verbal output.
- **Multimodal Cueing:** Effective intervention relies on a "triple-cue" approach:
 - **Visual:** Directing focus through imagery and observation.
 - **Audio:** Using tonal and verbal prompts.
 - **Gestural:** Incorporating body language to bridge communication gaps.
- **Tactile Sensorial Therapy:** The discussion touched on the necessity of sensory feedback. By using physical touch or tactile stimulants, therapists can help patients "re-map" the motor movements required for speech.

Gaps Identified in the Current System

1. The Aphasia Barrier (Expressive vs. Receptive) The most pressing clinical challenge is the disconnect between comprehension and expression. Stroke survivors frequently grapple with **aphasia**, a condition where the cognitive ability to understand speech remains intact, but the motor or neurological ability to construct words is lost. This creates an "internal isolation" where thoughts and emotions cannot be externalized.

2. Linguistic Exclusion A significant "Language Accessibility Gap" exists within the current market. Most assistive technologies are optimized for **English**, leaving those who speak regional languages—specifically **Kannada**—without viable digital support. For these patients, the lack of native-language tools is a direct barrier to effective rehabilitation.

3. The Burden of Manual Intervention The current model of care is heavily reliant on one-on-one manual therapy. This **high-touch dependency** creates a bottleneck in the healthcare system, as it exhausts therapist resources and limits the frequency of care available to the broader patient population.

4. Fragmented Digital Solutions While some digital aids exist, they are often one-dimensional. There is a noticeable lack of **holistic, integrated platforms** that can simultaneously offer visual demonstrations, audio feedback, and gesture-based cues to guide a patient through a complete exercise.

Project Status

At this stage, the definitive **problem statement** for the major project remains under deliberation. Having concluded two comprehensive site visits, the team is now synthesizing clinical insights to evaluate various intervention strategies in **assistive communication** and **rehabilitative technology**. Our next steps involve a triangulated approach—combining literature reviews, data synthesis, and continued expert dialogue—to finalize a high-impact research trajectory and solidify the project’s technical scope.

Conclusion

The follow-up visit to **AIISH** was instrumental in deepening our understanding of the recovery landscape for stroke survivors and those with speech deficits. The consultation underscored that successful patient outcomes are heavily dependent on **multimodal communication cues**, specialized therapy techniques, and the availability of inclusive digital platforms. These professional insights have provided a clear framework that will now anchor the research and design phases of our major project.